

MOXIFCIN EYE DROPS 0.5% W/V

Ingredient(s):

Each ml contains:
Moxifloxacin Hydrochloride (eq. to Moxifloxacin 5mg/ml) 5.45mg

Pharmacodynamic:

Moxifloxacin has *in vitro* activity against a wide range of Gram-positive and Gram-negative microorganisms. Moxifloxacin inhibits the topoisomerase II (DNA gyrase) and topoisomerase IV required for bacterial DNA replication, transcription, repair and recombination. The C8-methoxy moiety of moxifloxacin also lessens the selection of resistant mutants of Gram-positive bacteria compared to the C8-H moiety found in older fluoroquinolones. Moxifloxacin's bulky C-7 substituent group interferes with the quinolone efflux pump mechanism of bacteria.

Moxifloxacin is often bactericidal at concentrations equal to or slightly greater than inhibitory concentrations.

Fluoroquinolones, including moxifloxacin, differ in chemical structure and mode of action from β -lactam antibiotics, macrolides and aminoglycosides, and therefore may be active against bacteria resistant to β -lactam antibiotics, macrolides and aminoglycosides. Therefore, organisms resistant to these drugs may be susceptible to moxifloxacin.

In vitro resistance to moxifloxacin develops slowly via multiple-step mutations and occurs at a general frequency between 10^{-9} to 10^{-11} for Gram-positive bacteria.

Moxifloxacin has been shown to be active against most strains of the following microorganisms, both *in vitro* and in clinical infections:

Gram-positive bacteria:

- Corynebacterium species*
- Microbacterium species*
- Micrococcus luteus* [including erythromycin, gentamicin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus aureus* [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus epidermidis* [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus haemolyticus* [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus hominis* [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus warneri* [including erythromycin resistant strains]
- Streptococcus mitis* [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]
- Streptococcus pneumoniae* [including penicillin, erythromycin, gentamicin, tetracycline and/or trimethoprim resistant strains]
- Streptococcus viridans* [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]

Gram-negative bacteria:

- Acinetobacter species*
- Haemophilus "alconae"* [including ampicillin resistant strains]
- Haemophilus influenzae* [including ampicillin resistant strains]
- Klebsiella pneumoniae*
- Moraxella catarrhalis*
- Pseudomonas aeruginosa*

Other microorganisms:

Chlamydia trachomatis
Moxifloxacin has been shown to be active *in vitro* against most strains of the following organisms; however, the clinical significance of these data is unknown.

Gram-positive bacteria:

- Arthrobacter species*
- Bacillus cereus*
- Bacillus thuringensis*
- Corynebacterium accolens*
- Corynebacterium amycolatum*
- Corynebacterium bovis*
- Corynebacterium macginleyi*
- Corynebacterium propinquum*
- Corynebacterium pseudodiphtheriticum*
- Enterococcus faecalis*
- Exiguobacterium species*
- Kocuria kristinae*
- Kocuria "lindaea"*
- Kocuria rhizophila*
- Listeria monocytogenes*
- Microbacterium "harmaniae"*
- Microbacterium "otitidis"*
- Rothia mucilaginosa*
- Staphylococcus arlettae*
- Staphylococcus capitis*
- Staphylococcus caprae*
- Staphylococcus cohnii*
- Staphylococcus lugdunensis*
- Staphylococcus pasteuri*
- Staphylococcus saprophyticus*
- Staphylococcus sciuri*
- Streptococcus agalactiae*
- Streptococcus "conjunctivae"*
- Streptococcus cristatus*
- Streptococcus dysgalactiae*
- Streptococcus mitis*
- Streptococcus Groups C, G and F*
- Streptococcus "ocularis"*
- Streptococcus oralis*
- Streptococcus parasanguinis*
- Streptococcus pyogenes*
- Streptococcus salivarius*
- Streptococcus sanguis*
- Streptococcus "schlechlii"*

Gram-negative bacteria:

- Achromobacter xylooxidans*
- Acinetobacter baumannii*
- Acinetobacter calcoaceticus*
- Acinetobacter johnsonii*
- Acinetobacter junii*
- Acinetobacter lwoffii*
- Acinetobacter "mumbaiae"*
- Acinetobacter schindleri*
- Acinetobacter ursingii*
- Aeromonas caviae*
- Chryseobacterium indologenes*
- Chryseobacterium species*
- Citrobacter freundii*
- Citrobacter koseri*
- Enterobacter aerogenes*
- Enterobacter cloacae*
- Enterobacter hormaechei*
- Escherichia coli*
- Klebsiella oxytoca*
- Moraxella osloensis*
- Morganella morganii*
- Neisseria gonorrhoeae*
- Pantoea agglomerans*
- Proteus mirabilis*
- Proteus vulgaris*
- Pseudomonas oryziabittans*
- Pseudomonas stutzeri*
- Serratia liquefaciens*
- Serratia marcescens*
- Stenotrophomonas maltophilia*

Anaerobic microorganisms:

- Clostridium perfringens*
- Fusobacterium species*
- Porphyromonas species*
- Prevotella species*
- Propionibacterium acnes*

Other microorganisms:

- Atypical *Mycobacterium*
- Chlamydia pneumoniae*
- Legionella pneumophila*
- Mycobacterium avium*
- Mycobacterium marinum*
- Mycoplasma pneumoniae*

Pharmacokinetic:

Following topical ocular administration of this product, moxifloxacin was absorbed into the systemic circulation. The mean steady-state C_{max} and AUC were 2.7 ng/mL and 41.9 ng·hr/mL, respectively. These exposure values are approximately 1,600 and 1,200 times lower than the mean C_{max} and AUC reported after well-tolerated therapeutic 400mg oral doses of moxifloxacin. The plasma half-life of moxifloxacin was estimated to be 13 hours.

Indication(s):

- a) Moxifcin Eye Drop is indicated for the treatment of patients 1 year of age and older with bacterial conjunctivitis caused by susceptible strains of the following organisms:

Gram- positive bacteria:

- Corynebacterium species**
- Microbacterium species*
- Micrococcus luteus** [including erythromycin, gentamicin, tetracycline, and/or trimethoprim resistant strains]
- Staphylococcus aureus* [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus epidermidis* [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus haemolyticus* [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus hominis** [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
- Staphylococcus warneri** [including erythromycin resistant strains]
- Streptococcus mitis** [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]
- Streptococcus pneumoniae* [including penicillin, erythromycin, gentamicin, tetracycline and/or trimethoprim resistant strains]
- Streptococcus viridians* [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]

Gram-negative bacteria:

- Acinetobacter species*
- Haemophilus "alconae"* [including ampicillin resistant strains]
- Haemophilus influenza* [including ampicillin resistant strains]
- Klebsiella pneumoniae**
- Moraxella catarrhalis**
- Pseudomonas aeruginosa**

Other Microorganisms:

- Chlamydia trachomatis*

**Efficacy for this organism was studied in fewer than 10 infections.*

b) Moxifcin Eye Drop is indicated for preoperative and postoperative sterilization (when prophylactic antibiotic treatment is required).

Dosage and Administration(s):
Instill one drop in the affected eye 3 times a day for 7 days.

Route of Administration:
Ophthalmic.

Contraindication:
Hypersensitivity to moxifloxacin, to other quinolones, or to any of the components in this medication.

Warning and Precautions:

a) For ocular use only. Not for injection. This product should not be injected subconjunctivally or introduced directly into the anterior chamber of the eye.

b) In patients receiving systemically administered quinolones, serious and occasionally fatal hypersensitivity (anaphylactic) reactions have been reported, some following the first dose. Some reactions were accompanied by cardiovascular collapse, loss of consciousness, angioedema (including laryngeal, pharyngeal or facial oedema), airway obstruction, dyspnoea, urticaria, and itching.

c) If an allergic reaction to this product occurs, discontinue use of the product. Serious acute hypersensitivity reactions may require immediate emergency treatment. Oxygen and airway management should be administered where clinically indicated.

d) As with other anti-infectives, prolonged use may result in overgrowth of non-susceptible organisms, including fungi. If superinfection occurs, discontinue use and institute alternative therapy.

e) Whenever clinical judgment dictates, the patient should be examined with the aid of magnification, such as slit-lamp biomicroscopy, and, where appropriate, fluorescein staining.

f) Tendon inflammation and rupture may occur with systemic fluoroquinolone therapy including moxifloxacin, particularly in elderly patients and in those treated concurrently with corticosteroids. Therefore, treatment with this product should be discontinued at the first sign of tendon inflammation.

g) Patients should be advised not to wear contact lenses if they have signs and symptoms of bacterial conjunctivitis.

h) **Pediatric Use:** The safety and effectiveness of Moxifloxacin in infants below 1 year of age have not been established. There is no evidence that the ophthalmic administration of Moxifloxacin has any effect on weight bearing joints, even though oral administration of some quinolones has been shown to cause arthropathy in immature animals.

i) **Geriatric Use:** No overall differences in safety and effectiveness have been observed between elderly and younger patients.

Information for Patients:

a) Avoid contaminating the applicator tip with material from the eye, fingers or other source.

b) Systemically administered quinolones including Moxifloxacin have been associated with hypersensitivity reactions, even following a single dose. Discontinue use immediately and contact your physician at the first sign of a rash or allergic reaction.

Effects on ability to drive and use machines:
Temporary blurred vision or other visual disturbances may affect the ability to drive or use machines.
If blurred vision occurs at application, the patient must wait until the vision clears before driving or using machinery.

Interaction with Other Medicaments(s):
Given the low systemic concentration of moxifloxacin following topical ocular administration of the medicinal product, drug interactions are unlikely to occur.

Pregnancy and Lactation:
Pregnancy
Teratogenic Effects
Pregnancy Category C
There are no or limited amount of data from the use of this product in pregnant women. However, no effects on pregnancy are anticipated since the systemic exposure to moxifloxacin from topical ocular application is negligible.
Since there are no adequate and well-controlled studies in pregnant women, the product should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Lactation
It is unknown whether moxifloxacin / metabolites are excreted in human milk. Animal studies have shown excretion of low levels in breast milk after oral administration of moxifloxacin. However, at therapeutic doses of moxifloxacin no effects on the suckling child are anticipated.

Side Effects:

System Organ Classification	Adverse reactions
Blood and lymphatic system disorders	<i>Rare:</i> haemoglobin decreased
Nervous system disorders	<i>Uncommon:</i> headache <i>Rare:</i> paresthesia
Eye disorders	<i>Common:</i> eye pain, eye irritation <i>Uncommon:</i> punctate keratitis, dry eye, conjunctival haemorrhage, ocular hyperaemia, eye pruritus, eyelid oedema, ocular discomfort <i>Rare:</i> corneal epithelium defect, corneal disorder, conjunctivitis, blepharitis, eye swelling, conjunctival oedema, vision blurred, visual acuity reduced, asthenopia, erythema of eyelid
Respiratory, thoracic and mediastinal disorders	<i>Rare:</i> nasal discomfort, pharyngolaryngeal pain, sensation of foreign body (throat)
Gastrointestinal disorders	<i>Uncommon:</i> dysgeusia <i>Rare:</i> vomiting
Hepatobiliary disorders	<i>Rare:</i> alanine aminotransferase increased, gammaglutamyltransferase increased

Symptoms and Treatment of Overdose:
No information is available on overdosage in humans. If a topical overdose of this product occurs, the eye(s) may be flushed with tap water.

Storage Condition:
Store at temperature below 30°C.

Shelf Life:
2 years from the date of manufacture.
Discard 1 month after opening.

Product description:
A clear and greenish-yellow solution.

Dosage form and packaging available:
Solution in plastic (LDPE) bottle of 5ml.



Manufacturer and Product Registration Holder:
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